

Mathematics for Biomedical Engineering

L2:

Differentiation

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- To understand the interpretation of the derivative in terms of rate of change;
- To calculate derivatives using the limit definition;
- To solve complex derivatives using specialised techniques of differentiation;
- To calculate the equation of the tangent/normal to a curve at a given point;
- To determine the maximum, minimum and inflection points of a function.

- 1 Differentiation
 - Interpretation of a Derivative
 - Calculation of Derivatives
 - Higher Derivatives

- 2 Techniques and Applications of Differentiation
 - The Product Rule and The Quotient Rule
 - The Chain Rule
 - Implicit Differentiation
 - Parametric Differentiation
 - Logarithmic Differentiation
 - Tangents and Normals
 - Maximum and Minimum Values of a Function

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Rate of change

In engineering we are often interested in the **rate at which some variable is changing**.

Rates of change may be positive (variable increasing), negative (variable decreasing) or zero (no change).

- y is increasing more rapidly for $0 < x < 2$ than $-5 < x < 0$; (the rate of change of y is positive and greater for $0 < x < 2$);
- y is decreasing for $2 < x < 3$ (negative rate of change);
- y is constant for $3 < x < 5$ (zero rate of change).

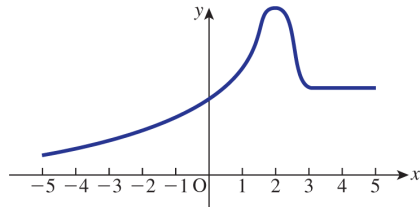


Figure: The function $y(x)$ changes at different rates for different values of x .

We calculate a *precise* value or expression for the rate of change of a function using **differentiation**.

Average rate of change across an interval

A function can have different rates of change at different points on its graphs.

The average rate of change across an interval is identical to the gradient of the chord across the interval.

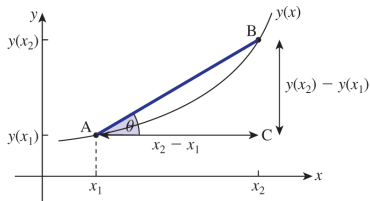


Figure: Average rate of change of y across the interval $x_1 \leq x \leq x_2$.

Average rate of change

$$\text{average rate of change of } y = \frac{y(x_2) - y(x_1)}{x_2 - x_1} \quad (1)$$

$$= \frac{BC}{AC} = \tan \theta \quad (2)$$

Average rate of change across an interval

Example: The voltage $v(t)$ across a capacitor varies with time, t , according to

$$v(t) = 3 + 2e^{-t}$$

Find the average rate of change of voltage as time varies

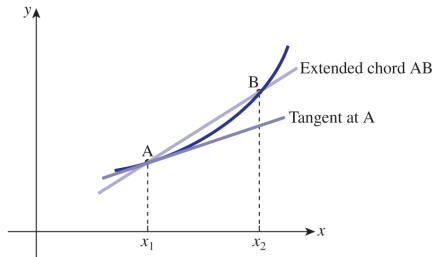
- from $t = 0$ to $t = 2$
- from $t = 1$ to $t = 3$

Rate of change at a point

In engineering we are often interested in the rate of change of a function at a point.

The gradient of the chord AB provides an approximation to the rate of change at A .

As B is moved closer to A , the gradient of the chord AB provides a *better* approximation to the rate of change at A .



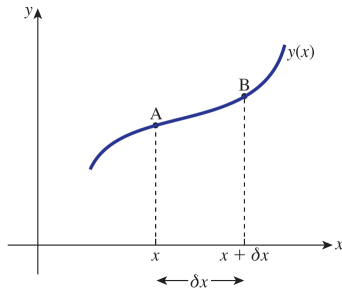
Rate of change at a point

Rate of change at a point A = gradient of tangent to the curve at A

Rate of change at a point

We can find the rate of change of a function $y(x)$ for any point x as follows.

$$\begin{aligned}
 \text{gradient of } AB &= \frac{\text{change in } y}{\text{change in } x} \\
 &= \frac{y(x + \delta x) - y(x)}{x + \delta x - x} \\
 &= \frac{\delta y}{\delta x}
 \end{aligned}$$



The gradient of AB gives the average rate of change of $y(x)$ from x to $x + \delta x$. As B approaches A , the distance δx decreases and the cord AB gets closer to the tangent at A .

Mathematical calculation of the rate of change of y at a point x

$$\text{Rate of change of } y(x) = \lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} \quad (3)$$

Rate of change at a point

Example: Find the rate of change of $y(x) = x^2$.

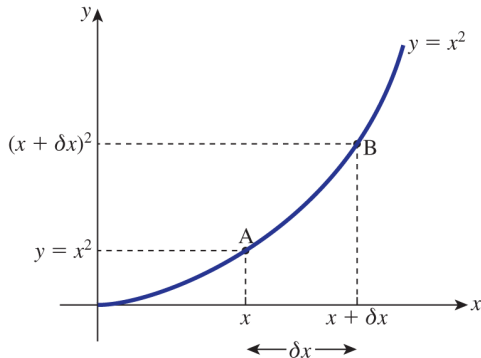


Figure: The gradient of the tangent at A is approximated by the gradient of the chord AB.

Derivative and Differentiation

The process of finding the rate of change of a given function is called **differentiation**.

The rate of change of a function is known as the **derivative** of the function.

Notation for the derivative of a function: $\frac{dy}{dx}$ or y'

When the independent variable is t , we can use the *dot notation*: $\frac{dz}{dt}$ or z' or $\dot{z}$

Mathematical calculation of the rate of change of y at a point x

$$\text{Rate of change of } y(x) = \lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \frac{dy}{dx} \quad (4)$$

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- Higher Derivatives

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Common functions and their derivatives

Function	Derivative
constant	0
x	1
kx	k
x^n	nx^{n-1}
kx^n	$k nx^{n-1}$
e^x	e^x
e^{kx}	ke^{kx}
$\ln x$	$\frac{1}{x}$
$\ln kx$	$\frac{1}{x}$
$\sin x$	$\cos x$
$\sin kx$	$k \cos kx$
$\sin(kx + \alpha)$	$k \cos(kx + \alpha)$
$\cos x$	$-\sin x$
$\cos kx$	$-k \sin kx$
$\cos(kx + \alpha)$	$-k \sin(kx + \alpha)$
$\tan x$	$\sec^2 x$
$\tan kx$	$k \sec^2 kx$
$\tan(kx + \alpha)$	$k \sec^2(kx + \alpha)$

k , n and α are constants, and all angles are in radians.

Common functions and their derivatives

$$\sinh x$$

$$\sinh kx$$

$$\cosh x$$

$$\cosh kx$$

$$\tanh x$$

$$\tanh kx$$

$$\sin^{-1} x$$

$$\sin^{-1} kx$$

$$\cos^{-1} x$$

$$\cos^{-1} kx$$

$$\tan^{-1} x$$

$$\tan^{-1} kx$$

$$\cosh x$$

$$k \cosh kx$$

$$\sinh x$$

$$k \sinh kx$$

$$\operatorname{sech}^2 x$$

$$k \operatorname{sech}^2 kx$$

$$\frac{1}{\sqrt{1-x^2}}$$

$$\frac{k}{\sqrt{1-k^2x^2}}$$

$$\frac{-1}{\sqrt{1-x^2}}$$

$$\frac{-k}{\sqrt{1-k^2x^2}}$$

$$\frac{1}{1+x^2}$$

$$\frac{k}{1+k^2x^2}$$

k , n and α are constants, and all angles are in radians.

Basic Rules

The following rules of the derivative enable us to extend the range of functions that we can differentiate.

Derivative of the sum or difference of two functions

The derivative of $f(x) \pm g(x)$ is

$$\frac{df}{dx} \pm \frac{dg}{dx} \quad (5)$$

Derivative of a function multiplied by a constant

The derivative of $kf(x)$ is

$$k \frac{df}{dx} \quad (6)$$

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Higher Derivatives

The function $\frac{dy}{dx}$ is more correctly called the **first derivative** of y .

By differentiating the first derivative, we obtain the **second derivative** of y ; by differentiating the second derivative, we obtain the **third derivative** of y and so on.

The second and subsequent derivatives are known as **higher derivatives**.

Notation for higher derivatives

If $y = y(x)$,

$$\text{first derivative} = \frac{dy}{dx} = y' \quad (7)$$

$$\text{second derivative} = \frac{d^2y}{dx^2} = y'' \quad (8)$$

$$\text{third derivative} = \frac{d^3y}{dx^3} = y''' \quad (9)$$

When calculating higher derivatives with respect to time, t , we can use the *dot notation*;

e.g. $\frac{d^2z}{dt^2} = \ddot{z}$

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The Product Rule and The Quotient Rule

The Product Rule

If

$$y(x) = u(x)v(x)$$

then

$$\frac{dy}{dx} = \frac{du}{dx}v + u\frac{dv}{dx} = u'v + uv' \quad (10)$$

Example: Differentiate $y = x^2 \sin x$

The Quotient Rule

If

$$y(x) = \frac{u(x)}{v(x)}$$

then

$$\frac{dy}{dx} = \frac{\frac{du}{dx}v - u\frac{dv}{dx}}{v^2} = \frac{u'v - uv'}{v^2} \quad (11)$$

Example: Differentiate $y = \frac{\sin x}{x}$

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The Chain Rule

The Chain Rule

If

$$y = y(z) \text{ and } z = z(x)$$

then

$$\frac{dy}{dx} = \frac{dy}{dz} \frac{dz}{dx} \quad (12)$$

Example: Given $y = (x^3 + x)^7$, find $\frac{dy}{dx}$

Chain Rule applied to $\ln$

If

$$y = \ln f(x)$$

then

$$\frac{dy}{dx} = \frac{f'}{f} \quad (13)$$

Example: Given $y = \ln(x^2 + x + 1)$, find $\frac{dy}{dx}$

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Implicit Differentiation

A function y is expressed **explicitly** when it is equated to an expression involving only x terms; i.e. it can be expressed in the form $y = f(x)$.

Examples: $y = x^2 + 3$, $y = \sin 2x$, $y = e^{3x} - 2x$

A function y is expressed **implicitly** when it is impossible to write it in the form $y = f(x)$. Instead, it can be defined by an equation of the form $f(x, y(x)) = 0$.

Examples: $x^2 + y^2 = 1$, $\sin x + y + e^x + e^{-y} = x^3 + y^3$

Implicit Differentiation

We can find $\frac{dy}{dx}$ by differentiating with respect to x each term of the equation $f(x, y(x)) = 0$ that defines the implicit function.

The y terms in the equation can be differentiated using the **chain rule**.

Rearranging terms will lead to $\frac{dy}{dx}$ expressed explicitly in terms of x and y .

Example: Find $\frac{dy}{dx}$ given $y^2 + x = x^3 + 3y$

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Parametric Differentiation

We use **parametric differentiation** when both the x and y variables (traditionally independent and dependent, respectively) depend on an **independent third variable**, t (usually thought of as 'time').

If t cannot be eliminated to write $y = f(x)$ (e.g. $y = \sin t + t^2$, $x = e^t + t$), then $\frac{dy}{dx}$ can be found using the following rule.

Parametric Differentiation

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{dt} \frac{dt}{dx} \\ &= \frac{dy}{dt} / \frac{dx}{dt}\end{aligned}$$

Example: Find $\frac{dy}{dx}$ given $y = \sin t + t^2$, $x = e^t + t$

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Logarithmic Differentiation

This is a useful technique when we need to differentiate $y = f(x)$ with $f(x)$ given by a cumbersome product. It is usually faster than using the product rule.

Technique

- 1 Take the natural logarithm of both sides of the given equation;
- 2 Apply the laws of logarithms to simplify the r.h.s.;
- 3 Differentiate both sides with respect to x ;
- 4 Solve for $\frac{dy}{dx}$.

Important results to use this technique

- $\frac{d}{dx}(\ln y) = \frac{1}{y} \frac{dy}{dx}$
- $\frac{d}{dx}(\ln f(x)) = \frac{1}{f} \frac{df}{dx}$
- $\ln e^x = x$
- $\ln e^{f(x)} = f(x)$
- $\ln AB = \ln A + \ln B$
- $\ln \frac{A}{B} = \ln A - \ln B$
- $\ln A^n = n \ln A$, with n a constant

Example: Find $\frac{dy}{dx}$ given $y = x^2(1+x)^3$

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Tangents and Normals

A **tangent** is a straight line that touches a curve in one point only. This point is referred to as the **point of contact**.

A **normal** is a straight line that is perpendicular to a tangent through the point of contact.

We have seen that the derivative also represents the gradient of a tangent to the curve defined by $y(x)$. We can use this fact when calculating the equations of a **tangent/normal to a curve**.

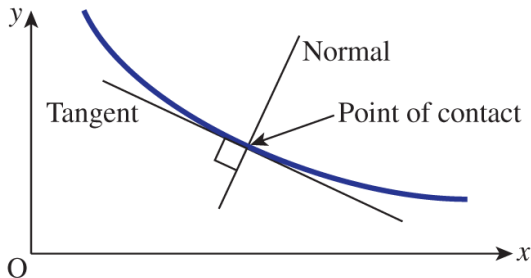


Figure: A tangent and normal to a curve, $y(x)$.

Calculating the equation of a tangent

The tangent has an equation of the form

$$t = m_1x + c_1$$

We calculate the gradient, m_1 , from $\frac{dy}{dx}$ and the constant, c_1 , from the coordinates of one point, usually the point of contact.

Example: Find the equation of the tangent to $y = x^2$ at the point $(2, 4)$.

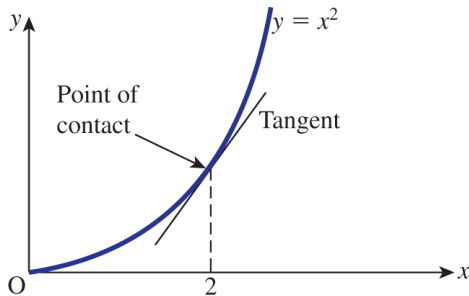


Figure: Tangent to the curve $y = x^2$ at the point $(2, 4)$.

Calculating the equation of a normal

The normal has an equation of the form

$$n = m_2x + c_2$$

Key Point

If two lines are perpendicular, then the product of their gradients is -1 .

Hence, if the tangent has gradient m_1 and the normal has gradient m_2 , then $m_1m_2 = -1$. So, once we know m_1 we can calculate m_2 . We find the constant, c_2 , from the coordinates of one point, usually the point of contact.

Example: Find the equation of the normal to $y = x^2$ at the point $(2, 4)$.

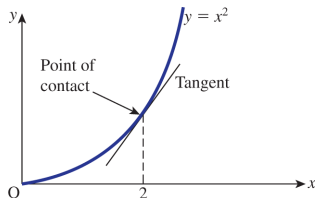


Figure: The normal to the curve $y = x^2$ at the point $(2, 4)$ is perpendicular to the tangent at the same point.

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Maximum and Minimum Values of a Function

Maximum and **minimum** points are located by examining the derivative of a function, rather than the function itself.

In the curves below, the point *A* is a **local maximum**; the point *B* is a **local minimum**. They are both **local extremes**; i.e. they are extreme values in their locality.

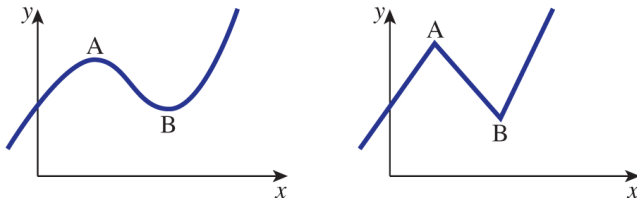


Figure: In both plots, *A* is a local maximum and *B* is a local minimum.

Key Point

At maximum and minimum points, $\frac{dy}{dx} = 0$ or $\frac{dy}{dx}$ does not exist.

Points at which $\frac{dy}{dx} = 0$ are known as **stationary points**.

The First Derivative Test

We can distinguish between a maximum point and a minimum point by considering y' on either side of the point.

First Derivative Test

It distinguishes between maximum and minimum points as follows:

- To the left of a maximum point, $\frac{dy}{dx}$ is positive; to the right $\frac{dy}{dx}$ is negative.
- To the left of a minimum point, $\frac{dy}{dx}$ is negative; to the right $\frac{dy}{dx}$ is positive.

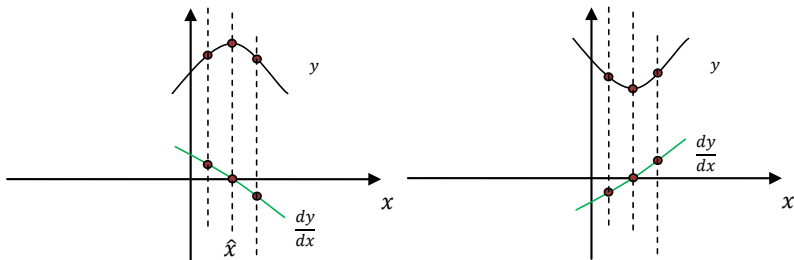


Figure: The first derivative decreases towards the right of a maximum point and increases towards the right of a minimum point.

The Second Derivative Test

We can also distinguish between a maximum point and a minimum point by considering the sign of y'' at these points.

Second Derivative Test

It distinguishes between maximum and minimum points as follows:

- If $y' = 0$ and $y'' < 0$ at a point, then the point is a maximum point.
- If $y' = 0$ and $y'' > 0$ at a point, then the point is a minimum point.
- If $y' = 0$ and $y'' = 0$ at a point, then the second-derivative test fails and we must return to the first-derivative test.

Points of Inflection

A point of inflexion is a point at which the concavity changes from concave up to concave down, or vice versa, as we move along the curve from left to right.

Concave up curve

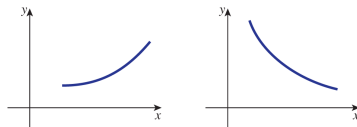


Figure: The gradient of both curves is increasing. Hence, y' is increasing and $y'' > 0$.

Concave down curve

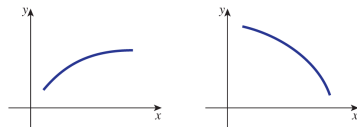


Figure: The gradient of both curves is decreasing. Hence, y' is decreasing and $y'' < 0$.

Condition for a point of inflection

At a point of inflection either $y'' = 0$ or y'' does not exist.

Points of Inflection

Condition for a point of inflection

At a point of inflection either $y'' = 0$ or y'' does not exist.

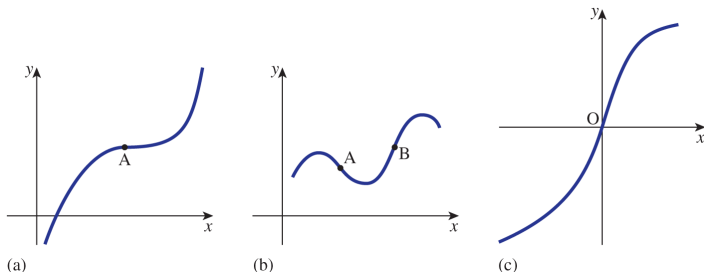


Figure: (a) There is a point of inflection at A; (b) there are points of inflection at A and B; (c) there is a point of inflection at O.

Important: Not all points where $y'' = 0$ or where y'' does not exist are points of inflection. However, we can **limit our search** to points where $y'' = 0$ or where y'' does not exist.

What We Learnt in Topic 2

- To understand the interpretation of the derivative in terms of rate of change;
- To calculate derivatives using the limit definition;
- To solve complex derivatives using specialised techniques of differentiation;
- To calculate the equation of the tangent/normal to a curve at a given point;
- To determine the maximum, minimum and inflection points of a function.